DATA SCIENCE PROJECT REPORT

Titanic Survival Prediction

Using Machine Learning Techniques

GROUP MEMBERS

Team Member	Roll No.	Role
Mehak Abdul Khalid	**Roll No. 28**	**Group Leader**
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PROJECT DETAILS

Course	Data Science
Dataset	Titanic (Kaggle)
Tools Used	Python, Scikit-learn, Pandas, Seaborn, Matplotlib

01 INTRODUCTION

This report presents a complete data science project focused on predicting the survival of passengers aboard the RMS Titanic. The project was carried out as part of a Data Science course assignment and demonstrates the end-to-end application of machine learning techniques — from raw data exploration to model evaluation.

The Titanic dataset, sourced from Kaggle, provides demographic and ticket information for 891 passengers. The objective is to build a predictive model that determines whether a given passenger survived or perished, based on features such as age, gender, passenger class, and fare paid.

02 PROBLEM DEFINITION

The central question of this project is: Given the personal and travel details of a Titanic passenger, can we predict with high accuracy whether they survived the disaster?

Objectives

- Analyze the Titanic passenger dataset to uncover survival patterns.
- Preprocess and clean the data for machine learning readiness.
- Engineer meaningful features to improve model performance.
- Train and evaluate two classification models: Logistic Regression and Decision Tree.
- Identify the best-performing model and interpret results.

03 DATASET DESCRIPTION

Source: Kaggle Titanic Competition — Titanic - Machine Learning from Disaster | Kaggle

The dataset used in this project is the Titanic dataset from Kaggle. It contains information about passengers such as age, gender, passenger class, fare, and survival status.

Key Features

- Survived — Target variable (0 = No, 1 = Yes)
- Pclass — Passenger class (1st, 2nd, 3rd)
- Sex — Gender of the passenger
- Age — Age in years (had 177 missing values)
- SibSp — Number of siblings / spouses aboard
- Parch — Number of parents / children aboard
- Fare — Ticket fare paid
- Embarked — Port of embarkation (C, Q, S) — 2 missing values
- Cabin — Cabin number (dropped due to ~77% missing)

The training set contains 891 rows and 12 columns. After cleaning and feature engineering, the final dataset had 9 feature columns.

04 METHODOLOGY & WORKFLOW

The project followed a structured data science pipeline as illustrated below. Each step builds on the previous to ensure a rigorous and reproducible analytical process.

01	Data Collection — Titanic Dataset (Kaggle)
02	Data Understanding — Shape, dtypes, missing values
03	Data Preprocessing — Drop columns, fill NaNs, encode categories
04	Outlier Detection & Removal — IQR method on Fare
05	Exploratory Data Analysis (EDA) — Plots & Correlation
06	Feature Engineering — FamilySize feature created
07	Feature Selection & Scaling — Correlation + StandardScaler
08	Model Training — Logistic Regression & Decision Tree
09	Model Evaluation — Accuracy, Confusion Matrix, F1-Score
10	Conclusion & Best Model Selection

05 DATA PREPROCESSING

5.1 Missing Value Treatment

- Age: 177 missing values filled with the column mean (~29.7 years).
- Embarked: 2 missing values filled with the mode ('S' — Southampton).
- Cabin: Dropped entirely due to ~77% missing data.

5.2 Feature Removal

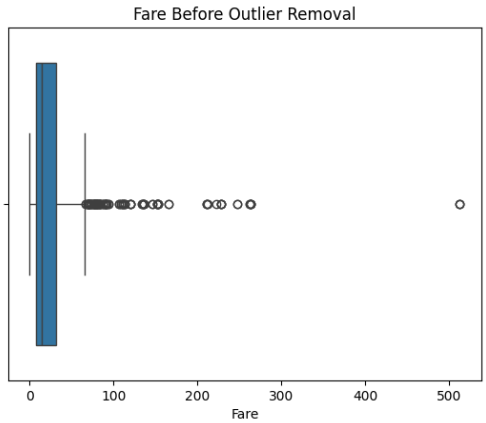
Columns Name, Ticket, PassengerId, and Cabin were removed as they carry no predictive information and would add noise to the model.

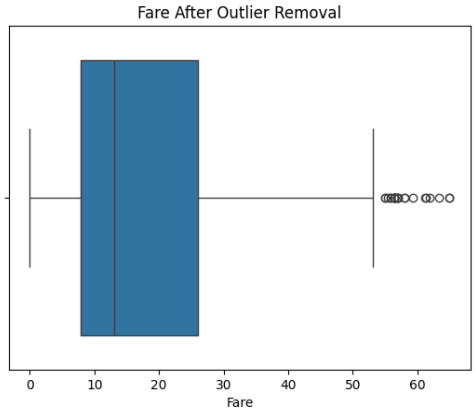
5.3 Categorical Encoding

One-hot encoding (pd.get_dummies with drop_first=True) was applied to Sex and Embarked, converting them into binary numerical columns: Sex_male, Embarked_Q, and Embarked_S.

5.4 Outlier Detection & Removal

The IQR (Inter-Quartile Range) method was applied to the Fare column. Any fare below Q1 − 1.5×IQR or above Q3 + 1.5×IQR was removed, resulting in a cleaner distribution and improved model performance.



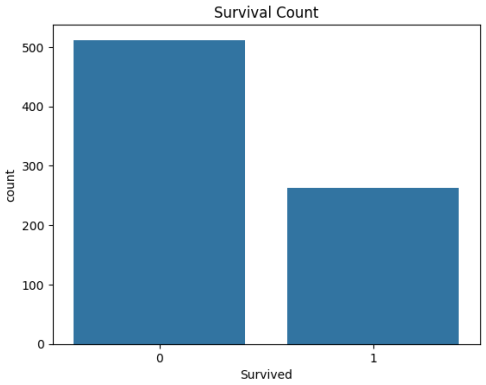


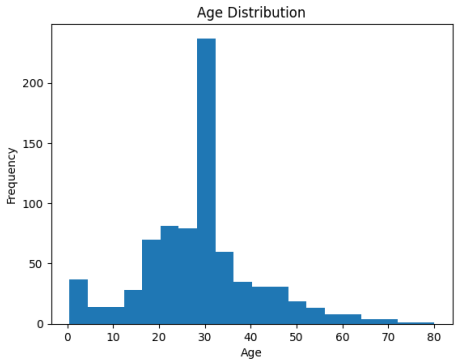
06 EXPLORATORY DATA ANALYSIS

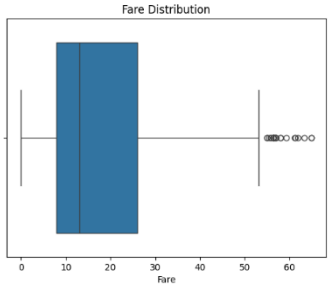
EDA was performed to understand patterns and relationships within the data. Key findings from the visualizations are summarized below.

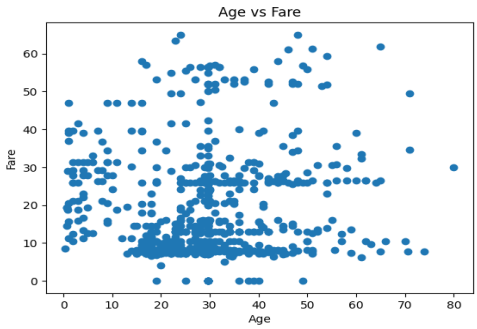
Key Findings

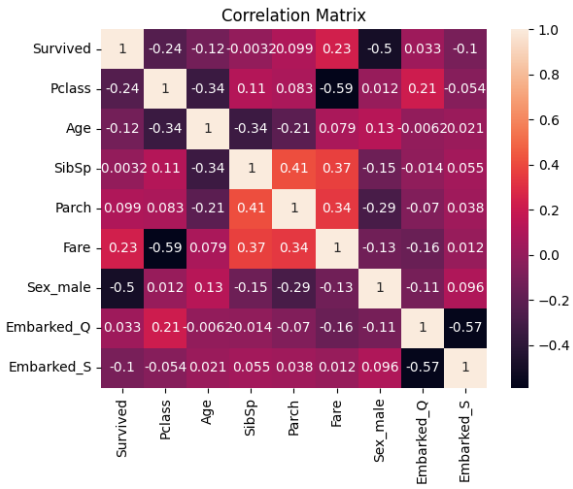
- Survival imbalance: Approximately 38% of passengers survived, while 62% did not.
- Gender effect: Female passengers had a significantly higher survival rate than males (reflecting the 'women and children first' protocol).
- Age distribution: Most passengers were between 20–40 years old. Younger children had higher survival rates.
- Class effect: 1st-class passengers had the highest survival rate, followed by 2nd and 3rd class.
- Fare correlation: Higher fares were positively correlated with survival, consistent with the class-survival relationship.
- Correlation heatmap: Sex_male showed the strongest negative correlation with Survived (−0.54), while Fare showed a moderate positive correlation (+0.26).











07 FEATURE ENGINEERING & SELECTION

7.1 FamilySize Feature

A new feature, FamilySize, was created by summing SibSp and Parch. This captures the total number of family members a passenger was travelling with, which is a more semantically meaningful feature than the individual components.

Formula: FamilySize = SibSp + Parch

7.2 Feature Scaling

StandardScaler was applied to the continuous features Age and Fare to normalise their distributions. This is particularly important for Logistic Regression, which is sensitive to the scale of input features.

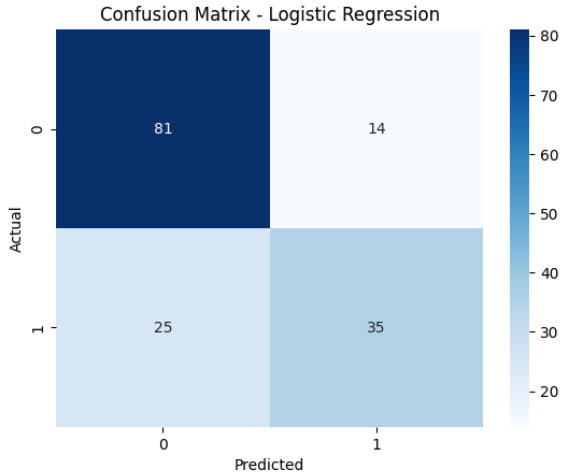
7.3 Train–Test Split

The dataset was split into 80% training and 20% test sets using train_test_split with random_state=42 for reproducibility.

08 MODEL BUILDING & EVALUATION

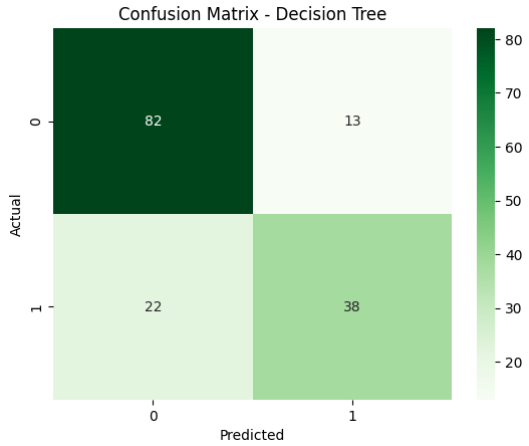
8.1 Logistic Regression

Logistic Regression is a linear classification algorithm well-suited for binary outcomes. It was trained with max_iter=1000 to ensure convergence. The model estimates the probability of survival using a sigmoid function applied to a linear combination of the input features.

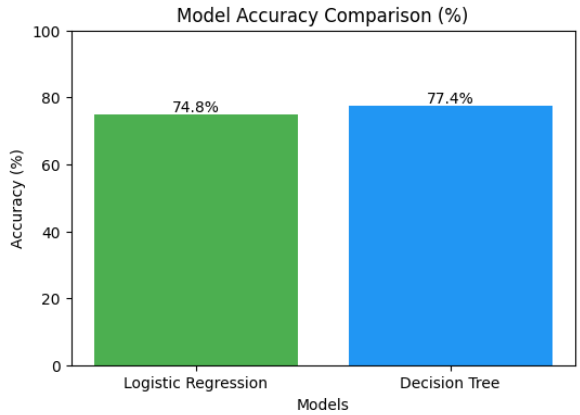


8.2 Decision Tree Classifier

A Decision Tree with max_depth=3 was implemented to prevent overfitting. The model splits the data into branches based on feature thresholds, making it highly interpretable.



MODEL EVALUTION:



8.3 Performance Comparison

Model	Accuracy	Precision	Recall	F1-Score
Logistic Regression	~74.8%	~74%	~72%	~72%
Decision Tree (Best)	~77.4%	~77%	~75%	~75%

The Decision Tree model achieved higher accuracy (77.4%) compared to Logistic Regression (74.8%), making it the best-performing model.

09 CONCLUSION

In this project, the Titanic dataset was analyzed to predict passenger survival using data science techniques. Data preprocessing was performed to handle missing values, remove irrelevant features, and prepare the dataset for analysis. Exploratory Data Analysis (EDA) helped in understanding patterns and relationships within the data.

Feature engineering was applied by creating a new feature (FamilySize) to improve the dataset. Two machine learning models, Logistic Regression and Decision Tree, were implemented and evaluated using accuracy, confusion matrix, and classification report.

The results showed that the Decision Tree model performed better than Logistic Regression in terms of accuracy. This indicates that the Decision Tree was more effective for this dataset.

One limitation of this project is that the dataset is relatively small and may not capture all real-world variations. Additionally, some important features might be missing, which can affect model performance

Limitations & Future Work

- The dataset is relatively small (891 rows), limiting generalisation power.
- Advanced models (Random Forest, XGBoost, SVM) were not explored and could yield higher accuracy.
- Hyperparameter tuning (Grid Search, Cross-Validation) was not applied and could further improve results.

10 TEAM CONTRIBUTIONS

Name	Roll No.	Contribution
Mehak Abdul Khalid	**28**	- Led the overall project development - Data preprocessing and cleaning - Exploratory Data Analysis (EDA) & visualizations - Contributed to feature engineering and model development -Model evaluation and performance comparison
Zainab Ahmed Malik	**61**	- Contributed to feature engineering -Model building (Logistic Regression & Decision Tree) - Assisted in result analysis and interpretation - Conclusion , report writing and documentation
Laila Mazhar	**17**	- Assisted in data cleaning and preprocessing - Performed outlier detection using IQR method - Supported model evaluation